

An Analysis of the Outcomes of Renta Dignidad in Bolivia

Diego Farfan

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Abstract

Bolivia's Renta Dignidad pays an unconditional cash transfer to every citizen aged 60 and older, regardless of income or contribution history. This paper uses the Encuesta de Hogares for 2021 through 2024 and a regression discontinuity design at the age-60 eligibility threshold to estimate the program's effects on non-labor income, employment, and poverty. Eligibility raises monthly non-labor income by about 308 bolivianos, most of it the transfer itself. Employment falls by 4 percentage points at the threshold, but the estimate is not statistically significant ($p = 0.160$), and an honest confidence interval that accounts for the discrete age variable runs from -10.2 to +1.0 points. The decline is concentrated among respondents with seven or more years of schooling, the group in which contributory pension recipients reduces the full-sample estimate to -1.8 points. Eligibility is associated with a 5.6 point fall in extreme poverty, but a simulation that removes Renta Dignidad income shows that this reduction is accounted for by the transfer itself. The program's clearest effect is on household income and poverty. The evidence that it induces retirement at age-60 is weak and difficult to separate from the contributory pension system.

Old-age transfers that do not depend on past contributions have spread across Latin America, Africa, and Asia. Since they raise income without requiring recipients to stop working, they lead to the question of whether unearned income draws older workers out of the labor force. Bolivia's Renta Dignidad is one of the most comprehensive of these types of programs, reaching almost every resident aged 60 and older. This paper estimates its effects on income, employment, and poverty at the age-60 eligibility threshold using pooled survey data from 2021 to 2024.

This paper draws on two groups of literature. The first concerns the labor supply effects of unearned income. Theory predicts that unearned income increases the demand for leisure, but empirical magnitudes vary widely. Imbens, Rubin, and Sacerdote (2001) provide a widely cited quasi-experimental estimate using lottery winners, finding that an additional dollar of unearned income reduces labor earnings by roughly 11 cents. Their results suggest that the size of a transfer relative to earnings matters for whether a measurable response appears.

The second group of literature concerns non-contributory social pensions in developing economies, where coverage is broad but benefit levels are low relative to earnings. This work generally finds that such pensions reduce elderly labor supply, although the size and form of the response depend on institutional context. De Carvalho Filho (2008) finds that the expansion of Brazil's rural pension in the early 1990s reduced the labor supply of elderly rural workers. Ardington, Case, and Hosegood (2009) show that South Africa's old age pension also affects the labor supply of other household members. Galiani, Gertler, and Bando (2016) find that Mexico's rural non-contributory pension shifted older adults away from paid work, with partial

substitution toward work in family activities. Where much employment is informal a transfer may change the composition of work without changing whether people work at all.

Renta Dignidad itself has been studied before. Escobar, Martinez, and Mendizabal (2013), the government's official evaluation, applies a regression discontinuity design at age-60 to a survey of households with older members and found poverty reductions of 11 to 14 percentage points at the cutoff, larger in urban than in rural areas. Hernani-Limarino and Mena (2015) exploit the reduction of the eligibility age from 65 to 60 when Renta Dignidad replaced Bonosol, using difference-in-differences and changes-in-changes models to estimate effects on welfare, savings, investment, and labor market outcomes. Borrella-Mas, Bosch, and Sartarelli (2019) use the age-60 discontinuity to study poverty, consumption, labor supply, and household composition, including the eligibility of spouses. This paper adds to that work in three ways. It uses more recent data, from 2021 through 2024, when the benefit was 350 bolivianos per month. It treats the contributory pension system as an explicit threat to identification, since the Prestacion Solidaria de Vejez creates a second age-related threshold at 58. Additionally, it uses the program's rule against drawing a salary to test a mechanism specific to salaried workers. Since age in the survey is measured in whole years, this paper also reports inference designed for discrete running variables.

Renta Dignidad was created by Law 3791 in November 2007 and first paid in February 2008, replacing Bonosol, which was paid starting at age 65 (Duran-Valverde and Barbero 2016). Its regulation, Supreme Decree 29400, originally set the benefit at 2,400 bolivianos per year for people without a contributory pension and 1,800 bolivianos for those with one. During the 2021

and 2024 period studied in this paper, recipients without a contributory pension received 350 bolivianos per month and pension recipients received 300, with the December payment doubled to include an aguinaldo (roughly translates to Christmas bonus). The annual benefit was therefore 4,550 bolivianos for non-pensioners and 3,900 for pensioners, equivalent to about 379 and 325 bolivianos per month (Gestora Publica, n.d.). Beginning in January 2026, Supreme Decree 5503 raised the monthly payment for recipients without contributory pensions by 150 bolivianos. The program costs roughly 1 percent of GDP and reaches about a third of Bolivian households (Duran-Valverde and Barbero 2016).

Renta Dignidad is often described as unconditional, but eligibility is not entirely independent of employment. According to the program administrator, beneficiaries must be Bolivian residents aged 60 or older who do not draw a salary from the public or private sector (Gestora Publica, n.d.). For most older Bolivians, who work on their own account or in family enterprises, the transfer generates only an income effect. For salaried workers, turning 60 also attaches a price to continued salaried work equal to the forgone benefit, which adds a substitution effect. Standard theory therefore predicts a reduction in labor supply at 60 through the income effect for everyone, and a larger reduction in salaried employment in particular.

Bolivia also operates a contributory pension system under Ley 065, enacted in December 2010, which governs the Sistema Integral de Pensiones (Ley 065 2010). The Prestacion Solidaria de Vejez becomes available from age 58 for workers with at least 120 months of contributions, and women with children may claim up to three years earlier. Since this threshold sits two years below Renta Dignidad's and because contributory coverage is concentrated among more

educated, formally employed workers, the two programs are difficult to separate in labor market outcomes.

The data is drawn from the Encuesta de Hogares published by Bolivia's Instituto Nacional de Estadística (INE) for 2021 through 2024. The four years are pooled and restricted to respondents aged 45 through 75, giving 40,015 observations. All estimates use INE's expansion factors revised to the 2025 population projections, which are consistent across the four survey years.

The primary outcome is employment, defined from INE's activity classification. A respondent is coded as employed if classified as "poblacion ocupada" meaning the respondent worked during the survey reference week. Respondents who are unemployed and searching are counted as not employed, so the measure does not distinguish job loss from labor force exit. Employment is further divided into salaried employment (wage employees, salaried employers, and domestic workers), self-employment and other work (own-account workers, non-salaried employers, cooperative members, and unpaid family workers), and public-sector employment which is identified from the type of employer.

The survey asks respondents aged 60 and older whether they receive Renta Dignidad and the monthly amount. Renta Dignidad income is constructed as the reported monthly amount multiplied by 13/12 to include the aguinaldo. Contributory pension receipt is measured from the reported amount of old-age pension, which the questionnaire asks respondents to report separately from Renta Dignidad. Poverty outcomes are INE's constructed indicators for extreme

and moderate income poverty and the poverty gap, together with per capita household income and income relative to the poverty line. Household composition variables are constructed by linking all members of each household.

Eligibility for Renta Dignidad is determined by a sharp age threshold, so its effect can be estimated by comparing respondents just above and below age 60. The estimating equation is:

$$Y_i = \alpha + \tau D_i + \beta_1 (Age_i - 60) + \beta_2 D_i (Age_i - 60) + \varepsilon_i$$

where Y_i is the outcome for an individual i , D_i equals 1 if age is at least 60, $Age_i - 60$ is the running variable centered at the threshold, and τ is the discontinuity at age 60. The equation is estimated by local linear regression with a triangular kernel and survey weights, separately on each side of the cutoff, within a bandwidth selected by the mean squared error criterion of Calonico, Cattaneo, and Titiunik (2014). For the main employment specification this bandwidth is 4.32 years. Point estimates are conventional, and confidence intervals and p-values use the robust bias-corrected procedure of the same authors, with standard errors clustered by primary sampling unit to reflect the survey design.

Age is recorded in completed years, so the running variable is discrete. Within the main bandwidth there are four distinct ages below the cutoff and five about it. Standard local polynomial methods assume a continuous running variable, and clustering standard errors by age performs poorly with so few support points (Kolesar and Rothe 2018). This paper therefore reports three complementary approaches. First, honest confidence intervals (Armstrong and Kolesar 2020) bound the curvature of the regression function and remain valid when the running variable is discrete. Second, a local randomization comparison contrasts weighted means at aged

59 and 60 alone (Cattaneo, Frandsen, and Titiunuk 2015). Third, a donut specification drops age 60, the bin in which some respondents have only just become eligible.

Since reported take-up is below 100 percent in the first year of eligibility, the sharp design estimates an intention-to-treat effect. A fuzzy regression discontinuity design uses reported receipt as the treatment. Placebo tests use only observations on the same side of the true cutoff, so that the discontinuity at 60 does not contaminate them.

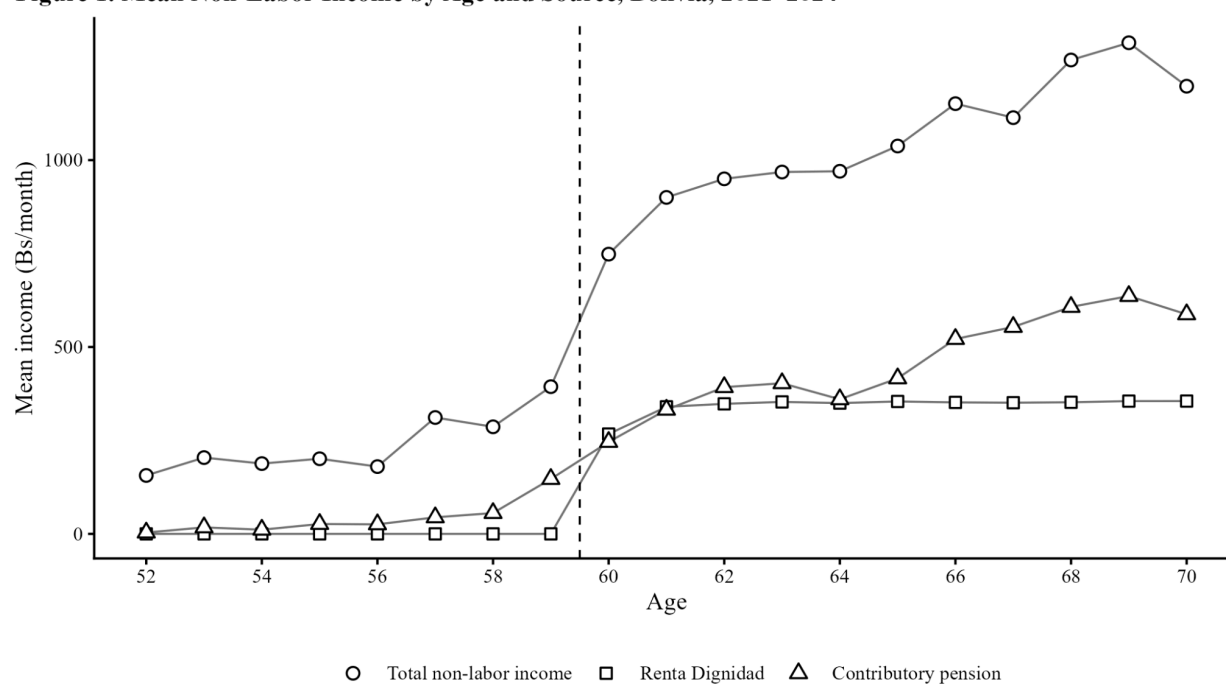
Figure 1 shows mean non-labor income by age, divided into Renta Dignidad income and contributory pension income. Table 1 reports the corresponding estimates. Non-labor income rises by 308 bolivianos per month at the threshold (95 percent confidence interval 49 to 515). Renta Dignidad income alone rises by 267 bolivianos, close to what first-year take-up and the benefit amount imply. Contributory pension income also rises across the threshold, by about 85 bolivianos, which is the reason as to why the total jump in non-labor income exceeds the transfer. The share of respondents reporting any non-labor income rises by 59 percentage points.

Table 1. First-Stage Effects of Renta Dignidad Eligibility

Outcome	Estimate	95% CI	p-value	Bandwidth	N
Non-labor income (Bs/month)	308.1	[49.0, 515.1]	0.018	3.30	9,404
Renta Dignidad income (Bs/month)	267.1	[255.9, 278.4]	<0.001	1.71	3,879
Contributory pension income (Bs/month)	84.8	[-39.0, 156.4]	0.239	4.54	11,978
Any non-labor income	0.586	[0.562, 0.624]	<0.001	1.95	3,879

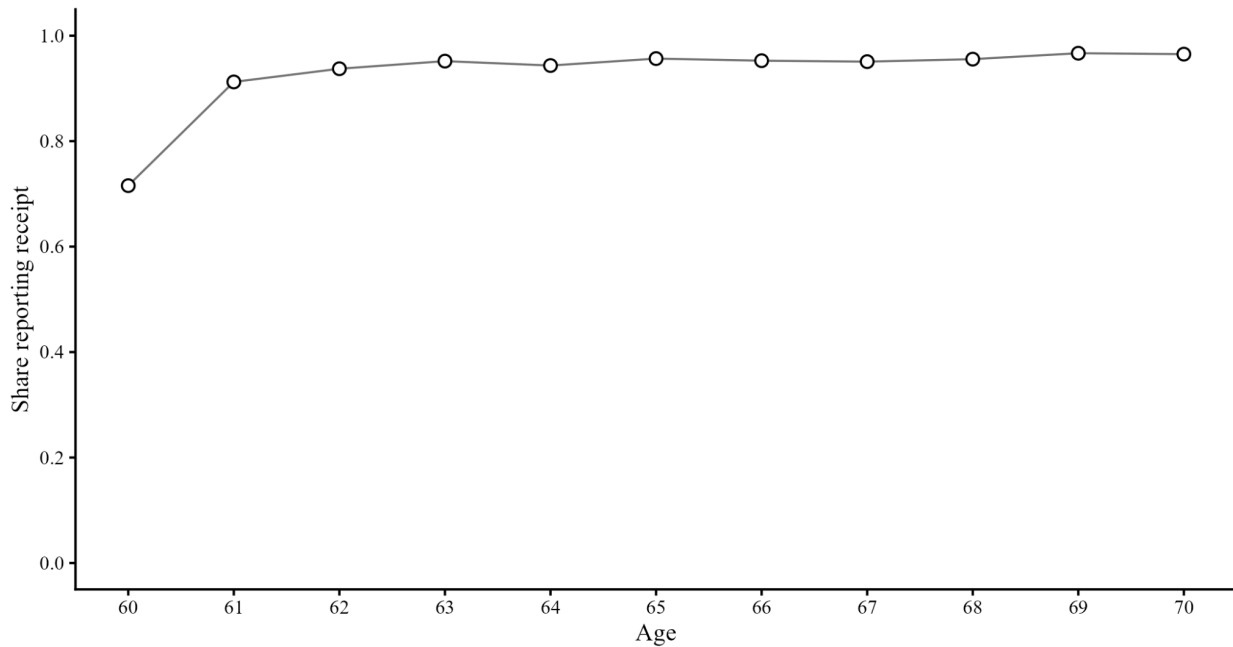
Local linear regression with triangular kernel and survey weights at the age-60 threshold. Point estimates are conventional; 95% confidence intervals and p-values are robust bias-corrected (Calonico, Cattaneo and Titiunuk 2014) with standard errors clustered by primary sampling unit. Each outcome uses its own MSE-optimal bandwidth.

Figure 1. Mean Non-Labor Income by Age and Source, Bolivia, 2021–2024



Note: Points are weighted means by single year of age. Renta Dignidad income is the reported monthly payment scaled by 13/12 to include the annual aguinaldo. The dashed line marks the age-60 eligibility threshold.

Reported receipt is 71.6 percent at age 60, 91.2 percent at 61, and 96 percent from age 63 onward (figure 2). Lower receipt at 60 is consistent with respondents who have only recently turned 60 along with administrative wait times. Among respondents aged 60 and older, receipt is higher in rural areas than in urban areas (96 percent against 91.7 percent) and higher among the poor than the non-poor (95.9 against 92.5 percent). Part of the gap in urban areas is likely due to the salary rule, since salaried work is more common there.

Figure 2. Reported Receipt of Renta Dignidad by Age, Bolivia, 2021–2024

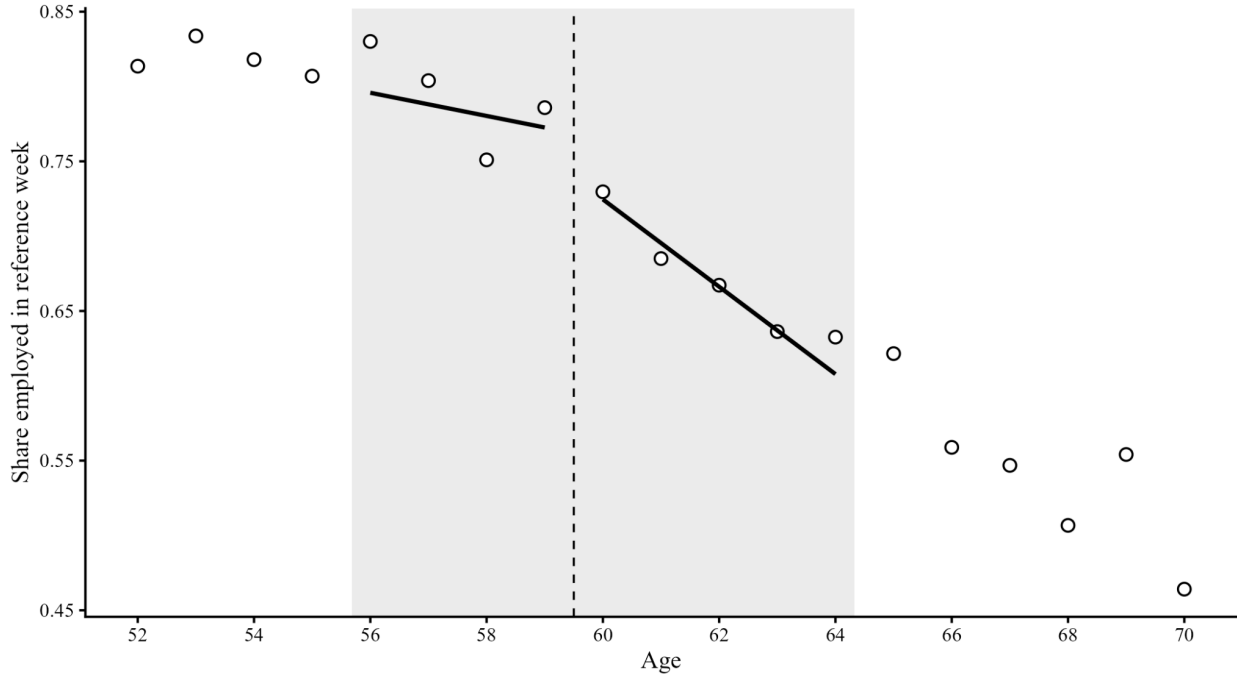
Note: Weighted share of respondents reporting receipt. The survey asks about receipt only for respondents aged 60 and older.

Table 2 reports regression discontinuity estimates of the effect of eligibility on employment. In the raw data, employment lies between 0.75 and 0.83 from age 54 through 59, then falls to 0.73 at 60 and 0.685 at 61, and declines further to 0.622 at 65 (Figure 3). The main estimate is a decline of 4 percentage points, with a robust confidence interval from -10.1 to +1.7 points and a p-value of 0.16. Adding year fixed effects leaves the estimate unchanged, while adding year, sex, schooling, area, and indigenous identification as covariates reduces it to -2.5 points ($p = 0.482$). The honest confidence interval runs from -10.2 to +1.0 points around an estimate of -4.6.

Table 2. Regression Discontinuity Estimates of the Effect of Eligibility on Employment

Specification	Estimate	95% CI	p-value	N
Main: local linear, MSE-optimal bandwidth	-0.040	[-0.101, 0.017]	0.160	11,978
Year fixed effects	-0.039	[-0.100, 0.018]	0.171	11,978
Covariates: year, sex, schooling, area, indigenous	-0.025	[-0.073, 0.034]	0.482	11,978
Honest CI (Armstrong-Kolesár bound)	-0.046	[-0.102, 0.010]	0.113	11,978
Local randomization, age 59 vs 60	-0.056	[-0.097, -0.016]	0.007	2,674
Donut: excluding age 60	-0.054	[-0.121, 0.008]	0.086	13,468
Threshold at age 61	-0.037	[-0.089, 0.030]	0.326	11,932
Fuzzy: receipt as treatment	-0.053	[-0.135, 0.021]	0.152	11,978
Excluding contributory pensioners	-0.018	[-0.078, 0.036]	0.468	10,971

Local linear regression with triangular kernel and survey weights at the age-60 threshold. Point estimates are conventional; 95% confidence intervals and p-values are robust bias-corrected (Calonico, Cattaneo and Titiunik 2014) with standard errors clustered by primary sampling unit. The honest interval (Armstrong and Kolesár 2020) bounds the second derivative of the regression function and is valid with a discrete running variable. Local randomization compares weighted means at ages 59 and 60. The fuzzy estimate uses reported receipt as the treatment. N counts observations within the bandwidth.

Figure 3. Share Employed by Age, Bolivia, 2021–2024

Note: Points are weighted means by single year of age. Solid lines are local linear fits with triangular kernel and survey weights, estimated separately on each side of the threshold within the MSE-optimal bandwidth of 4.3 years (shaded).

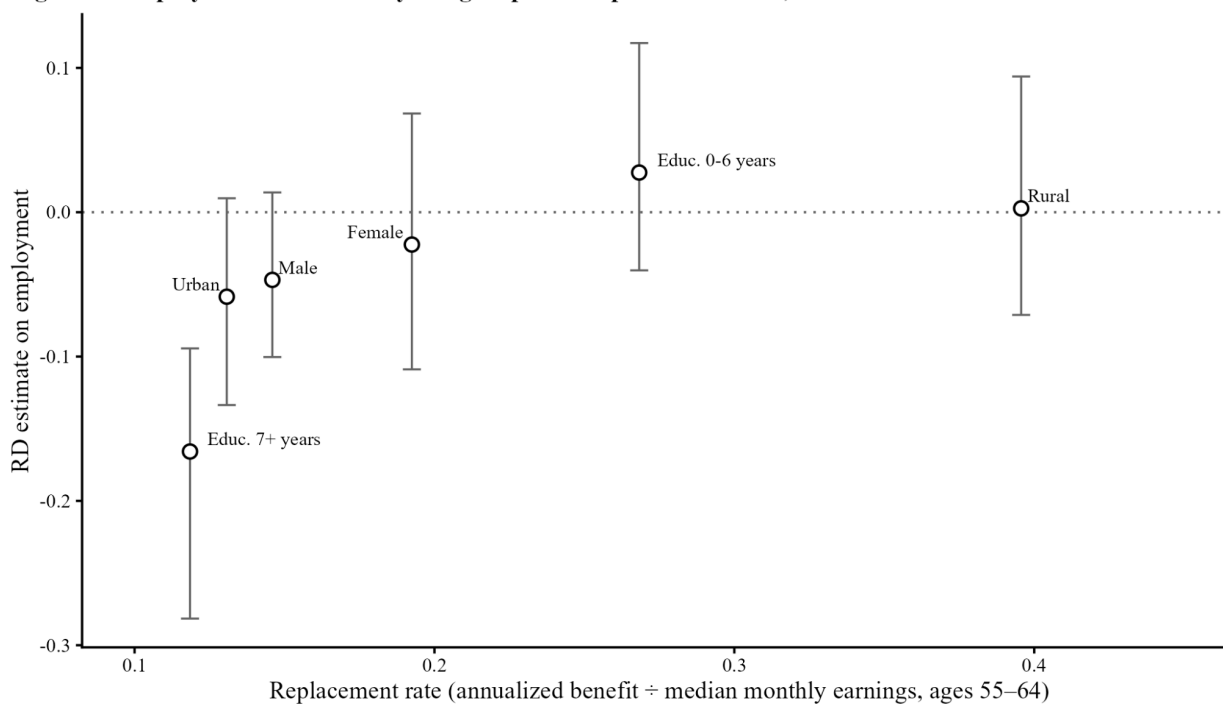
The local randomization comparison of ages 59 and 60 yields -5.6 points with a p-value of 0.007. This comparison assumes that employment would be flat between 59 and 60 without the program. Employment declines by roughly one point per year of age below the cutoff, so part of this difference reflects the ordinary age gradient rather than the program. The donut specification yields -5.4 points ($p = 0.086$). The fuzzy estimate is -5.3 points ($p = 0.152$). Moving the threshold to 61, where take-up exceeds 90 percent, yields -3.7 points ($p = 0.326$), so the estimate does not grow when take-up is nearly complete. Taken together, the estimates consistently point to a decline of about 4 to 5 points, but only the specification that ignores the age trend is statistically distinguishable from zero.

One explanation for a modest response is that the transfer is small relative to earnings. The annualized benefit of 379 bolivianos is 15.9 percent of median monthly labor income among employed respondents aged 55 to 64, and 9 percent of that group earns less than the benefit. If the response operated through the income effect, groups for whom the transfer replaces a larger share of earnings should show larger declines. Table 3 and Figure 4 report estimates by subgroup alongside each group's replacement rate. The largest decline, 16.6 points, is among respondents with seven or more years of schooling, whose replacement rate is 0.12, while respondents with six or fewer years, whose replacement rate is 0.27, show no decline. The difference between the two education groups is 19.3 points and is statistically significant ($p = 0.002$). The urban and rural estimates (-5.9 and +0.3 points) and the male and female estimates (-4.7 and -2.2 points) follow the same direction, but neither difference is statistically significant ($p = 0.273$ and $p = 0.649$).

Table 3. Employment Estimates and Replacement Rates by Subgroup

Subgroup	Median earnings	Replacement rate	RD estimate	95% CI	p-value
Education, 7+ years	3,200	0.118	-0.166	[-0.282, -0.094]	<0.001
Education, 0-6 years	1,413	0.268	0.027	[-0.040, 0.117]	0.338
Urban	2,900	0.131	-0.059	[-0.134, 0.010]	0.090
Rural	958	0.396	0.003	[-0.071, 0.094]	0.786
Male	2,598	0.146	-0.047	[-0.100, 0.014]	0.136
Female	1,970	0.192	-0.022	[-0.109, 0.068]	0.654

Median monthly labor earnings among employed respondents aged 55 to 64. Replacement rate is the annualized benefit of Bs 379 divided by median earnings. p-values for the difference within each pair: education 0.002; area 0.273; sex 0.649. Local linear regression with triangular kernel and survey weights at the age-60 threshold. Point estimates are conventional; 95% confidence intervals and p-values are robust bias-corrected (Calonico, Cattaneo and Titiunik 2014) with standard errors clustered by primary sampling unit.

Figure 4. Employment Estimates by Subgroup and Replacement Rate, 2021–2024

Note: Points are local linear RD estimates with 95% robust bias-corrected confidence intervals clustered by primary sampling unit. The benefit is Bs 4,550 per year (Bs 379 per month).

The subgroup pattern is difficult to reconcile with the income effect of the transfer, but it matches the distribution of contributory pensions. Table 4 and Figure 5 report contributory pension receipt by age and education. Among respondents with seven or more years of

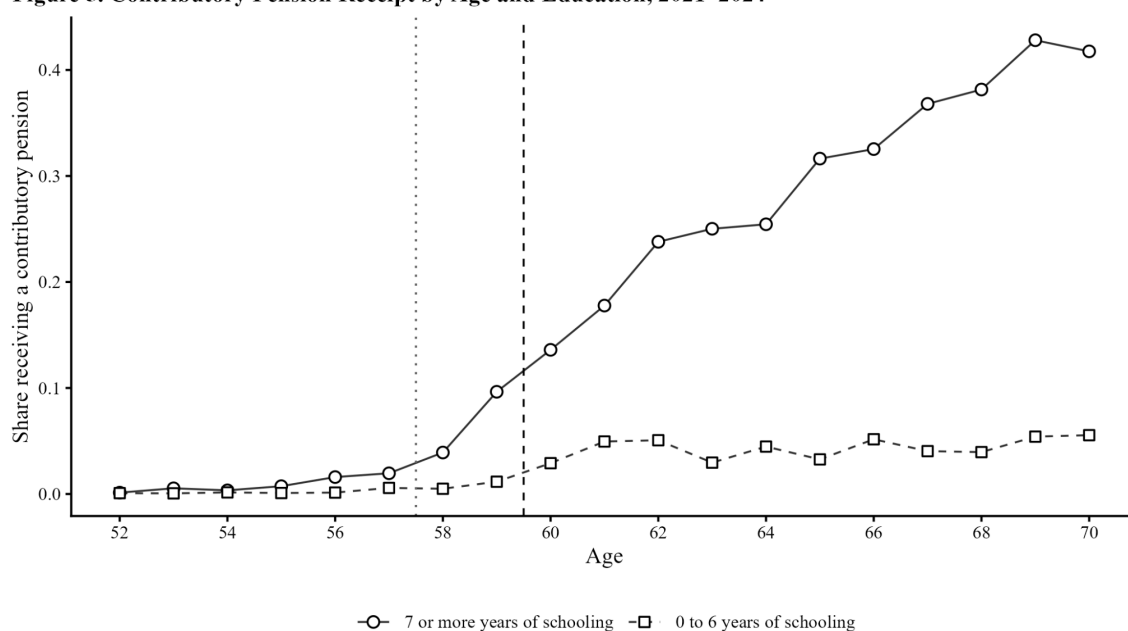
schooling, receipt rises from 0.02 at age 57 to 0.039 at 58, 0.096 at 59, 0.136 at 60, and 0.238 at 62. Among respondents with six or fewer years, it rises only from 0.006 to 0.051 over the same range. Receipt increases continuously from the statutory threshold at 58 rather than jumping at 60, but it does so steeply through the Renta Dignidad threshold and almost entirely within one education group. A local linear specification fitted to four or five ages on each side cannot fully separate this rising trend from a discontinuity. Applied to pension receipt itself among the more educated group, it returns a small negative estimate (-0.019 , $p = 0.285$) in a series that is visibly increasing across the cutoff.

Table 4. Share Receiving a Contributory Pension by Age and Education

Years of schooling	56	57	58	59	60	61	62	63
7 or more	0.016	0.020	0.039	0.096	0.136	0.178	0.238	0.250
0 to 6	0.001	0.006	0.005	0.012	0.029	0.050	0.051	0.030

Weighted shares reporting a contributory old-age pension (jubilación), excluding Renta Dignidad.

Figure 5. Contributory Pension Receipt by Age and Education, 2021–2024



Note: Weighted shares. The dotted line marks the Prestación Solidaria de Vejez threshold at 58; the dashed line marks Renta Dignidad eligibility at 60.

Table 5 reports employment estimates with and without respondents who receive a contributory pension. For the full sample, the estimate falls from -4 to -1.8 points, a reduction of 56 percent. For the more educated group, it falls from -16.6 to -10 points but remains statistically significant ($p = 0.003$). The estimate for the less educated group, whose contributory coverage is low, is essentially unchanged. Since receiving a contributory pension is itself related to leaving work, excluding pension recipients conditions the sample on an outcome, and the restricted estimates should be read as bounds rather than point estimates. What they show is that a large share of the employment decline at 60 occurs among people with contributory pension histories, and that a residual decline remains among the more educated even after pensioners are removed.

Table 5. Employment Estimates Excluding Contributory Pension Recipients

Sample	Unrestricted	Excluding pension recipients	Change
All respondents	-0.040 ($p = 0.160$)	-0.018 ($p = 0.468$)	-56%
Education, 7+ years	-0.166 ($p < 0.001$)	-0.100 ($p = 0.003$)	-40%
Education, 0-6 years	0.027 ($p = 0.338$)	0.035 ($p = 0.284$)	+28%

Because pension receipt is itself related to labor-market exit, restricted estimates condition on an outcome and should be read as bounds. Difference between education groups in the unrestricted estimate: -0.193 ($p = 0.002$). Local linear regression with triangular kernel and survey weights at the age-60 threshold. Point estimates are conventional; 95% confidence intervals and p-values are robust bias-corrected (Calonico, Cattaneo and Titiunik 2014) with standard errors clustered by primary sampling unit.

If the rule against drawing a salary bound in practice, salaried employment should fall at 60 even where self-employment does not. Table 6 and Figure 6 show no such pattern. The share of the population in salaried employment is 0.181 at both 59 and 60, and the estimated discontinuity is -0.2 points ($p = 0.818$). Public-sector employment, where the rule is most clearly enforced, shows no decline at 60 (+1.6 points, $p = 0.224$). Salaried employment does

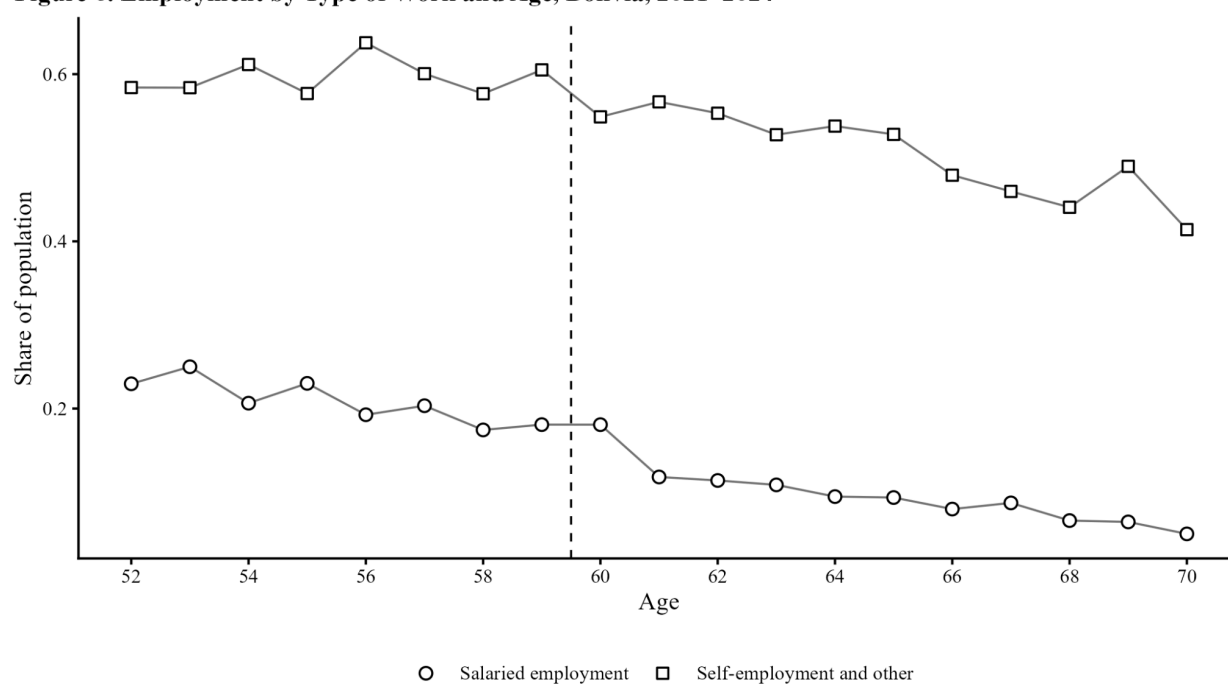
not fall sharply between 60 and 61, from 0.181 to 0.118, which coincides with rising contributory pension receipt and with the completion of Renta Dignidad take-up. This later drop is suggestive but cannot be attributed to either program with this design. Self-employment falls by approximately 2.9 points at 60 ($p = 0.380$). The data therefore do not support the salary rule as the mechanism behind the employment decline at 60.

Table 6. Discontinuities in Employment by Type of Work

Outcome	Estimate	95% CI	p-value	N
Employed (any)	-0.040	[-0.101, 0.017]	0.160	11,978
Salaried employment	-0.002	[-0.071, 0.056]	0.818	9,404
Self-employment and other	-0.029	[-0.093, 0.036]	0.380	11,978
Public-sector employment	0.016	[-0.013, 0.055]	0.224	9,404
Salaried, excluding pensioners	0.011	[-0.058, 0.073]	0.824	8,595

Outcomes are shares of the full population aged 45 to 75. Salaried employment includes wage employees, salaried employers and domestic workers; public-sector employment includes public administration and public enterprises. Local linear regression with triangular kernel and survey weights at the age-60 threshold. Point estimates are conventional; 95% confidence intervals and p-values are robust bias-corrected (Calonico, Cattaneo and Titiunik 2014) with standard errors clustered by primary sampling unit.

Figure 6. Employment by Type of Work and Age, Bolivia, 2021–2024



Note: Salaried employment includes wage employees, salaried employers and domestic workers; self-employment and other includes own-account workers, non-salaried employers, cooperative members and unpaid family workers.

Table 7 reports validity checks. Placebo cutoffs on employment return estimates between -5.5 and +3.0 points, none of them statistically significant at the 5 percent level. The largest placebo estimate below the cutoff is at 58 (-5.4 points, $p = 0.062$), which coincides with the contributory pension threshold. Placebo cutoffs on extreme poverty return estimates between -0.8 and +2.6 points, with p-values from 0.228 to 0.932.

Table 7. Validity Checks

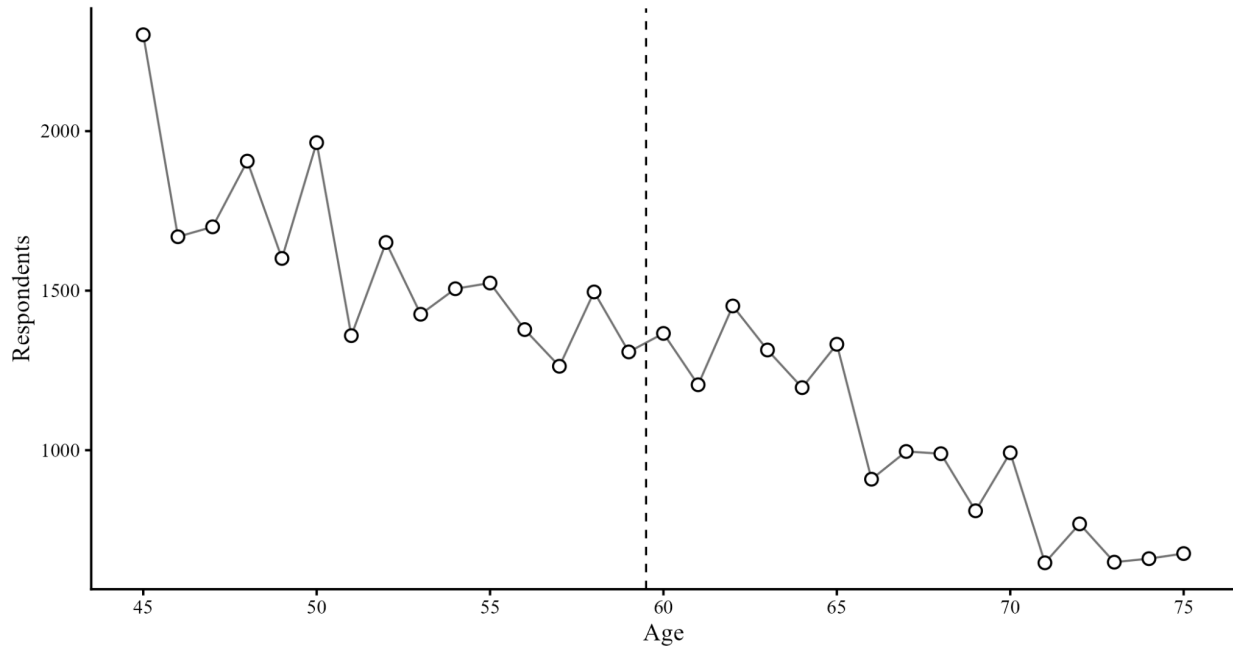
Test	Estimate	95% CI	p-value
Placebo, employment at 54	−0.017	[−0.068, 0.034]	0.537
Placebo, employment at 56	0.030	[−0.022, 0.083]	0.276
Placebo, employment at 58	−0.054	[−0.111, 0.003]	0.062
Placebo, employment at 62	0.023	[−0.055, 0.100]	0.566
Placebo, employment at 64	0.028	[−0.035, 0.091]	0.397
Placebo, employment at 66	−0.055	[−0.121, 0.010]	0.100
Placebo, extreme poverty at 54	0.002	[−0.041, 0.045]	0.919
Placebo, extreme poverty at 56	0.005	[−0.039, 0.049]	0.822
Placebo, extreme poverty at 58	0.016	[−0.027, 0.060]	0.467
Placebo, extreme poverty at 62	0.026	[−0.016, 0.069]	0.228
Placebo, extreme poverty at 64	−0.008	[−0.044, 0.029]	0.716
Placebo, extreme poverty at 66	−0.002	[−0.037, 0.034]	0.932
Balance: female	0.040	[−0.026, 0.129]	0.190
Balance: years of schooling	0.257	[−0.434, 1.433]	0.294
Balance: rural residence	−0.067	[−0.195, −0.003]	0.043
Balance: indigenous identification	−0.028	[−0.101, 0.031]	0.300
Density test at 60 (rddensity)	—	—	0.145
Binomial test, counts at ages 59 and 60	—	—	0.270

Placebo estimates use only observations on the same side of the true cutoff (ages below 60 for placebos at 54, 56 and 58; ages 60 and above for 62, 64 and 66) with honest confidence intervals, using the smoothness bound and bandwidth from the main honest estimate for each outcome. Balance tests use the main specification.

Of four predetermined covariates, three are balanced at the threshold. The share of rural residents falls by 6.7 points ($p = 0.043$). Since rural residents are more likely to work, this imbalance biases the unadjusted employment estimate downward, which is consistent with the smaller covariate-adjusted estimate in Table 2. The density test does not reject continuity in the number of respondents at 60 ($p = 0.145$), and a binomial test comparing counts at ages 59 and

60 gives $p = 0.270$. Figure 7 shows the number of respondents by age. Counts are elevated at ages ending in 0 and 5, as is common in household surveys, but the excess at 60 (9 percent above the average of its neighbors) is small compared with 50, 65, and 70 (27 to 36 percent) and similar to other even ages such as 58 and 62.

Figure 7. Number of Respondents by Single Year of Age, Bolivia, 2021–2024



Note: Unweighted counts in the pooled sample. Density test robust p-value = 0.145; binomial test of ages 59 and 60, $p = 0.270$.

Table 8 reports the employment estimate across bandwidths and polynomial orders. Local linear estimates range from -3.6 to -5.9 points for bandwidths between 3.5 and 10.5 years, and the honest confidence interval excludes zero only at the narrowest bandwidth. A local quadratic specification returns -18 points. With four or five distinct ages on each side, a second-order polynomial is fitted to very few points, so this estimate illustrates how fragile the employment result is to functional form rather than providing an alternative estimate. The local linear specification is retained.

Table 8. Sensitivity of the Employment Estimate to Bandwidth and Polynomial Order

Specification	Estimate	95% CI	N
Local linear, h = 3.5 (honest CI)	-0.059	[-0.116, -0.001]	9,404
Local linear, h = 4.5 (honest CI)	-0.038	[-0.094, 0.019]	11,978
Local linear, h = 5.5 (honest CI)	-0.036	[-0.097, 0.025]	14,834
Local linear, h = 6.5 (honest CI)	-0.039	[-0.107, 0.029]	17,249
Local linear, h = 7.5 (honest CI)	-0.041	[-0.117, 0.036]	19,671
Local linear, h = 10.5 (honest CI)	-0.049	[-0.163, 0.065]	27,436
Local quadratic, MSE-optimal bandwidth	-0.180	[-0.365, -0.058]	9,404

Local linear rows report honest confidence intervals with a common smoothness bound. The local quadratic row reports a robust bias-corrected interval; with four or five distinct ages on each side, it is shown to illustrate fragility rather than as an alternative estimate.

The poverty outcomes were examined after the primary employment analysis and were not specified in advance. Since they were selected with knowledge of the employment result and the analysis was not pre-registered, they should be interpreted as exploratory rather than confirmatory evidence. Table 9 reports the estimates. Extreme poverty falls by 5.6 percentage points at the threshold, income poverty by 8.4 points, and the poverty gap by 4.3 points, while log per capita household income rises by 0.22. In the estimation sample, extreme poverty falls from 12.2 percent at ages 56 to 59 to 6 percent at ages 60 to 64 (Figure 8)

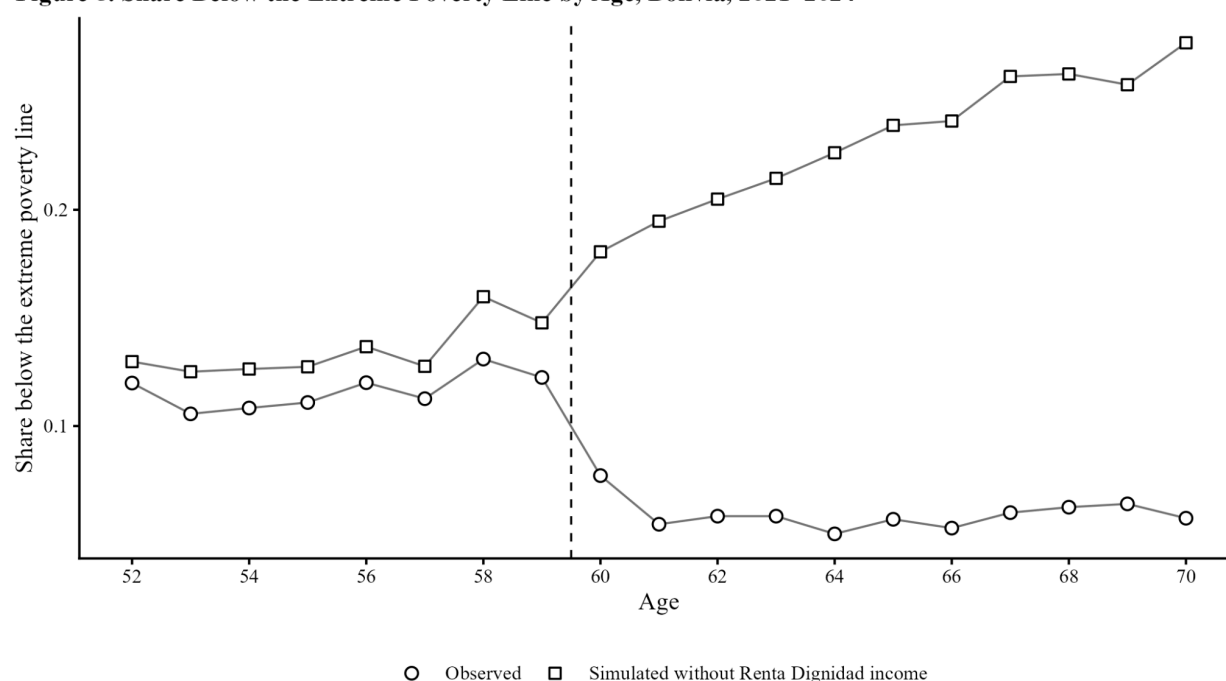
Table 9. Regression Discontinuity Estimates of the Effect of Eligibility on Poverty and Income

Outcome	Estimate	95% CI	p-value	N
Extreme poverty	-0.056	[-0.098, -0.007]	0.023	11,968
Income poverty	-0.084	[-0.186, -0.019]	0.016	9,397
Poverty gap	-0.043	[-0.078, -0.009]	0.013	11,968
Income relative to poverty line	0.298	[0.041, 0.636]	0.026	9,397
Log per capita household income	0.215	[0.080, 0.394]	0.003	9,396
Extreme poverty, simulated without Renta Dignidad	0.022	[-0.026, 0.063]	0.421	14,821
Income poverty, simulated without Renta Dignidad	-0.022	[-0.131, 0.042]	0.310	9,397

Poverty indicators are INE's constructed measures based on household per capita income, which includes Renta Dignidad. Simulated outcomes subtract all Renta Dignidad income in the household before comparing per capita income with the poverty lines. Local linear regression with triangular kernel and survey weights at the age-60 threshold. Point estimates are conventional; 95% confidence intervals and p-values are robust bias-corrected (Calonico, Cattaneo and Titiunik 2014) with standard errors clustered by primary sampling unit.

These poverty measures are computed from household per capita income, which includes Renta Dignidad, so a decline at 60 is partly an accounting consequence of the transfer. The relevant question is whether the reduction is offset by behavioral responses, such as lost labor income or reduced support from family. To answer it, the analysis subtracts all Renta Dignidad income received in each household and recomputes poverty status. In this simulated series, extreme poverty does not fall at 60. The estimated discontinuity at +2.2 points ($p = 0.421$), and the simulated income poverty estimate is -2.2 points ($p = 0.310$). Figure 8 shows the two series. Without the transfer, extreme poverty would rise gradually with age after 60 instead of falling. The observed poverty reduction is then accounted for by the transfer itself, and there is no evidence of an offsetting behavioral response.

Figure 8. Share Below the Extreme Poverty Line by Age, Bolivia, 2021–2024



Note: The simulated series subtracts all Renta Dignidad income received in the household from household income before comparing per capita income with the extreme poverty line.

The magnitudes are consistent with the transfer's scale. The annualized benefit of 379 bolivianos equals 85 percent of the average extreme poverty line and 44 percent of the average moderate poverty line in the estimation sample. Since poverty is measured per capita, the effect on any one household depends on its size. Households of respondents aged 60 to 64 average 2.4 members, so a single transfer raises per capita income by about 158 bolivianos. Among recipients aged 60 to 64, 24.8 percent remain income poor and 5.8 percent remain extremely poor.

Table 10 reports poverty estimates by area. Extreme poverty falls by 9.6 points in rural areas and by 2.4 points in urban areas, consistent with the transfer representing a larger share of income in rural households. The rural estimate is imprecise ($p = 0.158$), however, the difference between areas is not statistically significant ($p = 0.204$). This ordering differs from

Escobar, Martinez, and Mendizabal (2013), who find larger poverty reductions in urban areas, although their data come from an earlier period and a different survey.

Table 10. Poverty Estimates by Area of Residence

Outcome	Estimate	95% CI	p-value	N
Extreme poverty, urban	-0.024	[-0.051, -0.002]	0.036	10,586
Extreme poverty, rural	-0.096	[-0.186, 0.030]	0.158	3,411
Income poverty, urban	-0.064	[-0.153, -0.003]	0.041	6,745
Income poverty, rural	-0.084	[-0.229, 0.046]	0.191	3,411

Difference between rural and urban extreme-poverty estimates: -0.072 ($p = 0.204$). Local linear regression with triangular kernel and survey weights at the age-60 threshold.

Point estimates are conventional; 95% confidence intervals and p-values are robust bias-corrected (Calonico, Cattaneo and Titiunik 2014) with standard errors clustered by primary sampling unit.

Cash transfers may change living arrangements or allow other household members to change their labor supply. Table 11 reports estimates for household composition using all respondents aged 45 to 75, with the respondent excluded from counts of prime-age members. Household size, the number of co-resident children under 15, the presence of other prime-age members, the number of employed prime-age members, and the number of members aged 50 to 70 all return small estimates with p-values above 0.45. No change in household composition or spillover onto the employment of other household members is detected.

Table 11. Effect of Eligibility on Household Composition

Outcome	Estimate	95% CI	p-value	N
Household size	0.056	[-0.132, 0.298]	0.451	11,975
Co-resident children under 15	-0.033	[-0.139, 0.079]	0.586	11,978
Any prime-age member present	-0.010	[-0.087, 0.073]	0.862	9,404
Number of employed prime-age members	-0.028	[-0.149, 0.127]	0.874	11,978
Number of members aged 50-70	0.023	[-0.053, 0.117]	0.457	9,404

Estimated on all respondents aged 45 to 75, with the respondent excluded from counts of prime-age members. Local linear regression with triangular kernel and survey weights at the age-60 threshold. Point estimates are conventional; 95% confidence intervals and p-values are robust bias-corrected (Calonico, Cattaneo and Titiunik 2014) with standard errors clustered by primary sampling unit.

Taken together, the estimates support a clear conclusion about income and poverty and a considerably weaker one about employment. Eligibility raises non-labor income sharply at the threshold, mostly through the transfer itself, and is associated with substantial reductions in extreme and moderate poverty. These estimates survive robust inference, placebo cutoffs on poverty which show no comparable discontinuities, and the simulation confirms that the reduction reflects the transfer rather than changes in behavior. Since the benefit equals 85 percent of the extreme poverty line but only 44 percent of the moderate line, the pattern of a large reduction in extreme poverty with a quarter of recipients still poor is what the program's scale would lead one to expect.

The employment evidence is inconclusive. There is a decline of about 4 points at age 60 that is consistent across specifications, but it is not statistically significant under any inference that accounts for the age gradient and the discrete running variable. The honest confidence interval, from -10.2 to +1.0 points, is consistent both with no effect and with a reduction of up to

10 points, so the estimate should not be read as evidence that the effect is zero. Three features prevent attributing the decline to Renta Dignidad with confidence. The contributory pension system creates a competing threshold at 58, and excluding contributory pension recipients reduces the full-sample estimate by 56 percent. The decline is concentrated among more educated workers, the opposite of what an income effect would predict. Additionally, salaried employment, which the program's eligibility rule should affect most directly, shows no discontinuity at 60. What the evidence supports is narrower than a claim of no effect. The decline in employment at the eligibility threshold cannot be confidently attributed to Renta Dignidad, and retirement in this age range is concentrated among workers with contributory histories.

Four limitations bound these conclusions. First, the estimates are local to age 60 and do not extend to older ages. Second, since everyone approaching 60 knows the benefit is coming, the design holds expected lifetime wealth roughly constant and identifies the effect of receiving the transfer, a cash-on-hand effect, rather than the full wealth effect of the program over a working life. Third, the benefit amount is the same for all recipients without a contributory pension, so the design cannot say whether a larger or smaller transfer would change retirement behavior. The 150 boliviano increase introduced in January 2026 creates this variation, and once the 2026 Encuesta de Hogares is released, a difference-in-discontinuities design comparing the jump at 60 before and after the increase could estimate the effect of benefit size. Fourth, the age variable is recorded only in whole years, which limits the precision of the design based on this survey.

Renta Dignidad clearly increases non-labor income and reduces poverty, especially extreme poverty, among older Bolivianos, and it does so without detectable offsetting changes in behavior or household composition. The evidence that it induces retirement is inconclusive, since the age-60 employment decline is imprecisely estimated, concentrated among workers with contributory pension histories, and absent in the salaried employment that the program's eligibility rule most directly targets. For policy, the program's most clearly identifiable effects are on economic well-being rather than on retirement behavior.

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